

## YELABUGA, Russian Federation

WMO: 317330

Lat: 48.8034N

Lon: 135.8740E

Elev: 58

StdP: 100.63

Time Zone: 10.00 (E10)

Period: 86-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |              |            |              | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|--------------|------------|--------------|-----------------------|------------|--------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |              | 1%         |              |                       |            |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB         | WS         | MCDB         | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-29.0 | (c)<br>-27.4 | (d)<br>-32.3                  | (e)<br>0.2 | (f)<br>-27.9 | (g)<br>-30.7 | (h)<br>0.2 | (i)<br>-26.8 | (j)<br>8.3            | (k)<br>-17.8 | (l)<br>7.2 | (m)<br>-18.2 | (n)<br>2.8            | (o)<br>160 | (p)<br>0.533 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 7.9                    | 29.0            | 22.3 | 27.6 | 21.5 | 26.4 | 20.8 | 23.7                | 27.1 | 22.7 | 26.0 | 21.9 | 25.0 | 2.6                  | 260  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 22.6                            | 17.4 | 25.5 | 21.7 | 16.5 | 24.6 | 20.8 | 15.6 | 23.8 | 71.5          | 27.5 | 67.9 | 26.1 | 64.5 | 25.0 | 26.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |         |                    |         | n-Year Return Period Values of Extreme Temperature |         |            |         |            |         |            |         |    |    |
|-------------------|---------|---------|----------------------------|---------|--------------------|---------|----------------------------------------------------|---------|------------|---------|------------|---------|------------|---------|----|----|
|                   |         |         | Mean                       |         | Standard Deviation |         | n=5 years                                          |         | n=10 years |         | n=20 years |         | n=50 years |         |    |    |
| 1%                | 2.5%    | 5%      | Min                        | Max     | Min                | Max     | Min                                                | Max     | Min        | Max     | Min        | Max     | Min        | Max     | DB | WB |
| (a) 8.4           | (b) 7.3 | (c) 6.2 | (e) N/A                    | (f) N/A | (g) N/A            | (h) N/A | (i) N/A                                            | (j) N/A | (k) N/A    | (l) N/A | (m) N/A    | (n) N/A | (o) N/A    | (p) N/A |    |    |
| (a) 8.4           | (b) 7.3 | (c) 6.2 | (e) N/A                    | (f) N/A | (g) N/A            | (h) N/A | (i) N/A                                            | (j) N/A | (k) N/A    | (l) N/A | (m) N/A    | (n) N/A | (o) N/A    | (p) N/A |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 2.6    | -19.8 | -16.0 | -6.0  | 4.7   | 12.6  | 18.2  | 21.7  | 20.1  | 14.4  | 4.9   | -7.6  | -17.6 |
|                                                                                  | DBStd       | 15.37  | 4.52  | 5.02  | 5.55  | 4.26  | 4.33  | 3.48  | 2.71  | 2.68  | 3.60  | 4.52  | 6.51  | 4.66  |
|                                                                                  | HDD10.0     | 3902   | 925   | 729   | 497   | 168   | 24    | 0     | 0     | 0     | 6     | 169   | 527   | 857   |
|                                                                                  | HDD18.3     | 5992   | 1183  | 962   | 756   | 408   | 182   | 46    | 8     | 14    | 125   | 417   | 777   | 1115  |
|                                                                                  | CDD10.0     | 1184   | 0     | 0     | 0     | 10    | 105   | 245   | 363   | 313   | 138   | 11    | 0     | 0     |
|                                                                                  | CDD18.3     | 231    | 0     | 0     | 0     | 0     | 4     | 41    | 112   | 68    | 7     | 0     | 0     | 0     |
|                                                                                  | CDH23.3     | 1306   | 0     | 0     | 0     | 3     | 55    | 283   | 634   | 305   | 26    | 0     | 0     | 0     |
|                                                                                  | CDH26.7     | 228    | 0     | 0     | 0     | 0     | 6     | 67    | 122   | 33    | 1     | 0     | 0     | 0     |
| Wind                                                                             | WSAvg       | 3.1    | 3.1   | 3.1   | 3.0   | 3.4   | 3.3   | 2.6   | 2.4   | 2.4   | 2.8   | 3.8   | 3.8   | 3.4   |
| Precipitation                                                                    | PrecAvg     | 639    | 15    | 15    | 25    | 36    | 61    | 69    | 122   | 130   | 81    | 44    | 29    | 21    |
|                                                                                  | PrecMax     | 912    | 36    | 44    | 62    | 93    | 141   | 171   | 239   | 252   | 185   | 96    | 62    | 49    |
|                                                                                  | PrecMin     | 417    | 1     | 1     | 2     | 7     | 19    | 4     | 19    | 33    | 20    | 5     | 5     | 4     |
|                                                                                  | PrecStd     | 121    | 10    | 9     | 17    | 19    | 29    | 40    | 59    | 53    | 39    | 24    | 17    | 13    |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | -7.2  | 0.9   | 9.9   | 22.5  | 27.2  | 31.0  | 31.2  | 29.0  | 26.0  | 21.5  | 8.7   | -3.2  |
|                                                                                  |             | MCWB   | -7.9  | -2.5  | 4.3   | 12.2  | 16.7  | 21.6  | 23.7  | 23.3  | 20.1  | 14.5  | 4.5   | -4.7  |
|                                                                                  | 2%          | DB     | -10.0 | -2.5  | 6.6   | 18.7  | 24.9  | 28.7  | 29.2  | 27.5  | 23.7  | 17.5  | 5.5   | -7.2  |
|                                                                                  |             | MCWB   | -11.0 | -4.5  | 2.2   | 10.0  | 14.7  | 20.8  | 22.9  | 22.3  | 18.6  | 11.9  | 2.8   | -8.4  |
|                                                                                  | 5%          | DB     | -11.7 | -5.5  | 4.0   | 15.1  | 22.6  | 26.7  | 28.0  | 26.3  | 21.9  | 14.7  | 2.9   | -9.3  |
|                                                                                  |             | MCWB   | -12.6 | -7.0  | 0.3   | 7.8   | 13.6  | 19.8  | 22.4  | 21.4  | 17.6  | 9.3   | 0.3   | -10.1 |
|                                                                                  | 10%         | DB     | -13.5 | -8.1  | 2.0   | 12.0  | 20.6  | 24.6  | 26.7  | 25.0  | 20.3  | 11.9  | 1.1   | -10.8 |
|                                                                                  |             | MCWB   | -14.2 | -9.4  | -0.8  | 5.7   | 13.1  | 19.2  | 21.7  | 20.7  | 16.6  | 7.7   | -0.4  | -11.6 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | -8.1  | -2.2  | 4.5   | 13.2  | 17.6  | 22.5  | 24.7  | 24.6  | 21.0  | 14.6  | 5.6   | -4.7  |
|                                                                                  |             | MCDB   | -7.1  | 0.3   | 9.6   | 20.9  | 24.2  | 28.4  | 27.8  | 27.3  | 23.5  | 19.8  | 8.2   | -3.1  |
|                                                                                  | 2%          | WB     | -10.9 | -4.5  | 2.4   | 10.1  | 16.2  | 21.4  | 24.1  | 23.4  | 19.7  | 12.2  | 3.0   | -8.3  |
|                                                                                  |             | MCDB   | -10.0 | -2.6  | 6.0   | 17.8  | 22.9  | 27.0  | 27.9  | 26.1  | 22.6  | 16.9  | 5.6   | -7.4  |
|                                                                                  | 5%          | WB     | -12.6 | -6.9  | 0.8   | 8.0   | 15.0  | 20.5  | 23.3  | 22.3  | 18.5  | 10.2  | 1.0   | -10.0 |
|                                                                                  |             | MCDB   | -11.9 | -5.6  | 3.5   | 14.3  | 20.4  | 25.5  | 26.8  | 25.0  | 21.1  | 13.7  | 2.3   | -9.3  |
|                                                                                  | 10%         | WB     | -14.2 | -9.3  | -0.6  | 6.4   | 13.7  | 19.6  | 22.5  | 21.4  | 17.1  | 8.2   | -0.3  | -11.5 |
|                                                                                  |             | MCDB   | -13.4 | -8.1  | 1.7   | 11.0  | 19.3  | 23.9  | 25.6  | 23.9  | 19.6  | 11.6  | 1.0   | -10.8 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 6.8   | 8.4   | 8.4   | 9.3   | 9.7   | 8.7   | 7.9   | 7.5   | 8.9   | 8.3   | 6.7   | 6.2   |
|                                                                                  |             | MCDBR  | 7.7   | 10.4  | 9.6   | 14.3  | 13.7  | 11.6  | 9.8   | 9.8   | 10.7  | 12.4  | 8.1   | 7.2   |
|                                                                                  |             | MCWBR  | 7.0   | 8.6   | 6.4   | 7.8   | 6.4   | 5.1   | 4.7   | 4.8   | 5.5   | 7.4   | 6.0   | 6.4   |
|                                                                                  | 5% WB       | MCDBR  | 7.6   | 10.0  | 8.7   | 13.5  | 11.4  | 10.0  | 8.6   | 7.9   | 9.1   | 11.4  | 7.3   | 7.1   |
|                                                                                  |             | MCWBR  | 6.9   | 8.3   | 5.9   | 7.6   | 5.9   | 4.8   | 4.7   | 4.6   | 5.2   | 7.2   | 5.5   | 6.4   |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.288  | 0.335 | 0.408 | 0.528 | 0.562 | 0.511 | 0.502 | 0.445 | 0.411 | 0.440 | 0.375 | 0.302 |       |
|                                                                                  | taud        | 2.076  | 1.943 | 1.824 | 1.751 | 1.772 | 1.997 | 2.067 | 2.236 | 2.237 | 1.973 | 2.002 | 2.051 |       |
|                                                                                  | Ebn at Noon | 740    | 777   | 775   | 720   | 717   | 763   | 764   | 794   | 781   | 648   | 606   | 653   |       |
|                                                                                  | Edh at Noon | 88     | 129   | 173   | 208   | 212   | 170   | 157   | 127   | 115   | 126   | 96    | 79    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.81   | 3.04  | 4.47  | 4.77  | 5.25  | 5.62  | 5.23  | 4.58  | 3.72  | 2.48  | 1.59  | 1.38  |       |
|                                                                                  | RadStd      | 0.07   | 0.10  | 0.28  | 0.47  | 0.44  | 0.68  | 0.46  | 0.31  | 0.28  | 0.18  | 0.12  | 0.08  |       |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (6 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page